

Unit 12, Lesson 4: Writing Algebraic Expressions from Descriptions of Real-World Contexts Equation

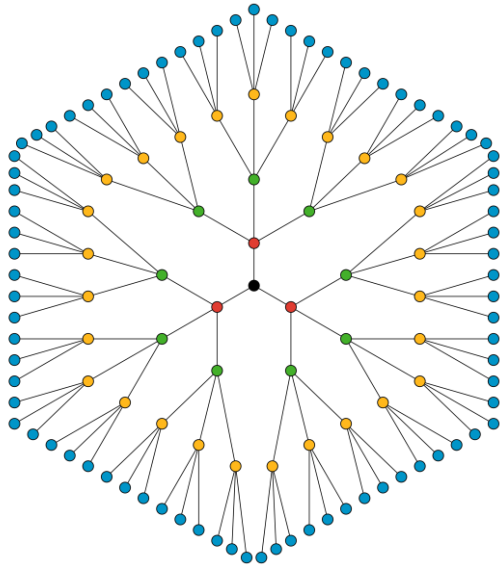
Benchmark

MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.

Learning Target

- ☐ I can translate a real-world description into an algebraic expression.

12.4.1 Warm-Up: Notice and Wonder



1. What do you notice about the image?
2. What do you wonder about the image?

12.4.2 Collaboration: Matching Activity

Part 1

Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	

A.

n

B.

x

x

x

x

x

x

x

x

x

x

C.

p

10

D.

w

10

Part 2

Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	
2. A number divided by ten	
3. The sum of a number and ten	
4. A number times ten	

12.4.3 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.
 - a. Write an algebraic expression that represents this situation.
 - b. What does the algebraic expression represent?
2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.
 - a. Write an algebraic expression to represent this situation.
 - b. What does the algebraic expression represent?
3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.
 - a. Write an algebraic expression to represent this situation.
 - b. What does the algebraic expression represent?

12.4.4 Collaboration: Error Analysis

Nero was working on a matching activity where he was given written descriptions for a series of mathematical expressions. Nero had to create an algebraic expression for each description.

1. Below are three written descriptions with the algebraic expressions Nero wrote. Determine whether each expression is correct. If the expression is incorrect, explain why it isn't correct and write the correct expression.

Written Description	Nero's Expression	Correct / Incorrect Explanation
André purchased the latest gaming bundle with x new add-ons, which is 7 times as many as his cousin Jaydah has.	$x + 7$	
Trenton volunteers with the Future Leaders of our Youth. He was helping to collect coats for 5 families that were in need this winter. Trenton collected x coats and wanted to break up the total number of coats equally among the families.	$x - 5$	
Elena created 2 more videos for her YouTube Channel. She originally had x videos posted.	$2 + x$	

12.4.5 Your Turn!

Write an algebraic expression to represent each situation.

1. Joe is 9 years younger than Drew, who is n years old.

2. Jenna's salary, s , is lowered by \$120.

3. Maria has 12 cupcakes and wants to split them among f friends.

4. Target has 8 cookie packages with c cookies in each.

12.4.6 Wrap-Up: Growth Mindset

1. Write a note to yourself about a mistake you learned from today.

**Means
I
Start
To
Acquire
Knowledge
Experience
Skills**

